

Evolutionary Computation and its Applications

Today's goals



Module 1

Fundamental Concepts in Optimization

Module 2

Genetic Algorithm

Module 3

Particle Swarm Optimization

Module 6 Introduction

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Introduction to Module 6's Study Video • 1 min

Module 6-1 Particle Swarm Optimization

✔

Particle Swarm Optimization: A First Introduction Video • 6 min

✔

Particle Swarm Optimization: Operators Video • 7 min

✔

Particle Swarm Optimization: Workflow Demo Video • 11 min

✔

Materials on Particle Swarm Optimization Reading • 10 min

✔

Understanding Particle Swarm Optimization Mechanics Reading • 5 min

Module 6-2 Use Particle Swarm Optimization 1

✔

Use PSO via MATLAB: PSO Function Video • 6 min

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Use PSO via MATLAB: Employ the Function Video • 3 min

✔

Materials on Using Particle Swarm Optimization Reading • 10 min

Module 6-3 Use Particle Swarm Optimization 2

✔

Use PSO via MATLAB: Case Studies Video • 6 min

✔

Materials on Using Particle Swarm Optimization Reading • 10 min

✔

Applying PSO to a Real Engineering System Reading • 5 min

Module 6 Quiz

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Module 6 Quiz Practice Assignment • 30 min

Module 6 Assignment

●

Module 6 Assignment Graded App Item • 30 min

●

Objective function for Assignment 6 Reading • 5 min

●

Designing and Applying Particle Swarm Optimization to Solve an Engineering Optimization Problem Dialogue • 5 min

●

Module 6 Recap Reading • 5 min

●

Keeping Learning with UofG Online Reading • 5 min

Applying PSO to a Real Engineering System



Reflect on the following questions:

- In the PID control example, why are K_pK_p , K_iK_i , and K_dK_d treated as **decision variables**, and not as constraints or objectives?
- Why is the **integral of error over time** a more meaningful objective than error at a single time instant?
- How does linking PSO with a **Simulink model** change the nature of the optimization problem compared to analytical benchmarks?
- What information do the **convergence plot**, **final objective value**, and **runtime** each provide about optimization success?
- Why does enabling **parallel computing** reduce runtime without changing the final solution?

This discussion helps you consolidate how PSO is used in practice when objective functions are expensive, simulation-based, and rooted in real engineering systems.

Go to next item

✔ Completed

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